

SHOFIA FAUZIYAH

Course by NordEdu
SQL #11

15 DAYS OF USING
SQL

Mastering Data Retrieval & Management with
PostgreSQL

Challenge

SYNTAX

```
Query Query History
1  -- Write a query that returns all payment details including
2  -- - the number of payments that were made by this customer
3  -- and that amount
4  -- Order the results by payment_id.
5  SELECT
6      p.payment_id,
7      p.customer_id,
8      p.staff_id,
9      p.rental_id,
10     p.amount,
11     p.payment_date,
12     COUNT(*) OVER (PARTITION BY p.customer_id, p.amount) AS no_payments_with_that_amount
13 FROM payment p
14 ORDER BY p.payment_id;
```

OUTPUT

Data Output

Messages

Notifications

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notes

Menampilkan detail informasi dari tabel payment. Selain itu, kita juga ingin tahu berapa kali seorang customer melakukan pembayaran dengan jumlah yang sama. Untuk menghitung hal itu, kita gunakan fungsi COUNT sebagai window function dengan PARTITION BY Hasil count dari PARTITION BY p.customer_id, p.amount lalu akan menampilkan angka 3 di kolom no_payments_with_that_amount, hasil query akan menampilkan detail pembayaran lengkap, informasi tambahan tentang berapa kali customer tersebut melakukan pembayaran dengan nominal yang sama

Challenge

notes

SYNTAX

```
Query Query History
1  -- Write a query that returns the running total of how late the flights are
2  -- (difference between actual_arrival and scheduled arrival) ordered by flight_id
3  -- including the departure airport.
4  -- As a second query, calculate the same running total but partition also by the
5  -- departure airport.
6  -- total keseluruhan
7  SELECT
8      flight_id,
9      departure_airport,
10     SUM(actual_arrival - scheduled_arrival)
11         OVER (ORDER BY flight_id) AS sum_running_total
12 FROM flights
13 ORDER BY flight_id;
14
15 -- total per departure airport
16 SELECT
17     flight_id,
18     departure_airport,
19     SUM(actual_arrival - scheduled_arrival)
20         OVER (PARTITION BY departure_airport ORDER BY flight_id) AS sum_running_total
21 FROM flights
22 ORDER BY departure_airport, flight_id;
```

Query ini dapat memudahkan kita sebagai user untuk menghitung running total keterlambatan penerbangan. Dengan running total, kita bisa melihat akumulasi keterlambatan dari waktu ke waktu. Total keseluruhan akan menunjukkan seberapa besar total keterlambatan yang sudah terjadi dari semua penerbangan.

- total keseluruhan → akumulasi semua penerbangan.
- total per departure airport → akumulasi khusus per bandara.

dengan ORDER BY membuat hasilnya berurutan, sedangkan PARTITION BY memisahkan perhitungan berdasarkan kelompok tertentu (dalam hal ini bandara keberangkatan)

OUTPUT total keseluruhan

Data Output Messages Notifications			
	flight_id [PK] integer	departure_airport character (3)	sum_running_total interval
1	1	DME	00:09:00
2	2	DME	00:10:00
3	3	DME	00:14:00
4	4	DME	00:14:00
5	5	DME	00:14:00
Total rows: 33121		Query complete 00:00:00.568	

OUTPUT total per departure airport

Data Output Messages Notifications			
	flight_id [PK] integer	departure_airport character (3)	sum_running_total interval
1	20981	AAQ	00:02:00
2	20982	AAQ	00:04:00
3	20983	AAQ	00:04:00
4	20984	AAQ	00:04:00
5	20985	AAQ	00:04:00
Total rows: 33121		Query complete 00:00:04.150	

Challenge

notes

SYNTAX

```
16 -- Write a query that returns the customers' name, the country and how many
17 -- payments they have. For that use the existing view customer_list.
18 -- Afterwards create a ranking of the top customers with most sales for each
19 -- country. Filter the results to only the top 3 customers per country.
20
21 -- Hitung jumlah pembayaran per customer
22 SELECT
23     cl.name,
24     cl.country,
25     COUNT(pm.payment_id) AS count
26 FROM customer_list cl
27 JOIN customer c ON cl.name = (c.first_name || ' ' || c.last_name)
28 JOIN payment pm ON c.customer_id = pm.customer_id
29 GROUP BY cl.name, cl.country
30 ORDER BY cl.country, count DESC;
31
32 -- Ranking top customer per negara (filter top 3)
33 SELECT *
34 FROM (
35     SELECT
36         cl.name,
37         cl.country,
38         COUNT(pm.payment_id) AS count,
39         RANK() OVER (PARTITION BY cl.country ORDER BY COUNT(pm.payment_id) DESC) AS rank_in_country
40     FROM customer_list cl
41     JOIN customer c ON cl.name = (c.first_name || ' ' || c.last_name)
42     JOIN payment pm ON c.customer_id = pm.customer_id
43     GROUP BY cl.name, cl.country
44 ) ranked
45 WHERE ranked.rank_in_country <= 3
46 ORDER BY ranked.country, ranked.rank_in_country;
```

Dengan query ini user dapat mengetahui berapa banyak pembayaran yang dilakukan setiap customer dan kemudian melihat siapa saja yang paling banyak melakukan transaksi di tiap negara.

Pertama count pembayaran per customerm, asilnya: nama, negara, dan total pembayaran.

Langkah kedua menambahkan ranking menggunakan RANK() dengan PARTITION BY country. Ini membuat perhitungan ranking terpisah per negara, Ini juga dapat mempermudah untuk menganalisis pelanggan paling aktif berdasarkan region

OUTPUT Hitung jumlah pembayaran per customer

Data Output				Messages	Notifications
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	name text	country text	count bigint		
1	VERA MCCOY	Afghanistan	18		
2	JUNE CARROLL	Algeria	37		
3	MARIO CHEATHAM	Algeria	28		
4	JUDY GRAY	Algeria	25		
5	ANTHONY SCHWAB	American Samoa	20		
Total rows: 599		Query complete 00:00:00.279			

OUTPUT Ranking top customer per negara (filter top 3)

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 238

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Page

	name text	country text	count bigint	rank_in_country bigint
1	VERA MCCOY	Afghanistan	18	1
2	JUNE CARROLL	Algeria	37	1
3	MARIO CHEATHAM	Algeria	28	2
4	JUDY GRAY	Algeria	25	3
5	ANTHONY SCHWAB	American Samoa	20	1

Total rows: 238

Query complete 00:00:00.241

THANKYOU